

CNuvS Ex. 6 - Jan Lindauer, Paul Feidieker

Task 4

- (a) 1. Segmentation, Addressing, Multiplexing
2. Connection Control
[1, p. 4]
- (b) The component 130.83.47.181 is the IP address representing the receiving host on the network, which a reverse DNS lookup identifies as a server belonging to TU Darmstadt (Hostname: cms-sip02.hrz.tu-darmstadt.de.). The component 443 is the port number representing the specific receiving service, which stands for HTTPS. [1, p. 5], [2]
- (c)
- (d) The major limitation of Alternating Bit Protocol is that it can only handle one packet at a time, which means that the sender has to wait for an acknowledgment for each packet before sending the next one. This leads to inefficient use of network resources, especially in high-latency networks, as the sender spends a lot of time waiting for acknowledgments instead of transmitting data. This becomes severe if the distance between the sender and receiver is large, leading to increased latency due to the longer propagation time and reduced throughput. [1, p. 8, 9]

Task 5

- (a)
- (b)
- (c)
- (d)

Task 6

- (a) Packet size in bits:

$$L = 13 \times 1024 \times 8 \text{ bit} = 106\,496 \text{ bit}$$

Data rate in bit/s:

$$R = 35 \times 2^{30} \text{ bit/s} = 37\,580\,963\,840 \text{ bit/s}$$

Total data in bits:

$$D = 3.7 \times 1024 \times 1024 \times 8 \text{ bit} = 31\,063\,040 \text{ bit}$$

Transmission time per packet:

$$t_{\text{tx}} = \frac{L}{R} = \frac{106\,496}{37\,580\,963\,840} \approx 2.8338 \times 10^{-3} \text{ ms} = 2.8338 \mu\text{s}$$

Number of Packets:

$$N = \left\lceil \frac{D}{L} \right\rceil = \left\lceil \frac{31\,063\,040}{106\,496} \right\rceil = \lceil 291.68... \rceil = 292 \text{ packets}$$

In the alternating bit protocol, the sender must wait for the ACK before sending the next packet.

One full round-trip cycle takes:

$$\begin{aligned}
t_{\text{cycle}} &= t_{\text{tx}}(\text{packet}) + t_{\text{prop}} + t_{\text{tx}}(\text{ACK}) + t_{\text{prop}} \\
&= 2 \cdot t_{\text{tx}} + 2 \cdot t_{\text{prop}} \\
&= 2 \times 2.8338 \mu\text{s} + 2 \times 84 \text{ ms} \\
&\approx 0.0057 \text{ ms} + 168 \text{ ms} \\
&\approx 168.0057 \text{ ms}
\end{aligned}$$

For the last packet we do not need to wait for the ACK to return:

$$\begin{aligned}
T_{\text{total}} &= (N - 1) \cdot t_{\text{cycle}} + t_{\text{tx}}(\text{packet}) + t_{\text{prop}} \\
&= 291 \times 168.0057 \text{ ms} + 2.8338 \mu\text{s} + 84 \text{ ms} \\
&= 48\,889.6 \text{ ms} + 84.0028 \text{ ms}
\end{aligned}$$

$$T_{\text{total}} \approx 48\,973.6 \text{ ms} \approx \mathbf{48.9736 \text{ s}}$$

- (b) The sender transmits all packets back-to-back without waiting for individual ACKs. Only one ACK is sent at the very end:

$$\begin{aligned}
T_{\text{total}} &= N \cdot t_{\text{tx}}(\text{packet}) + t_{\text{prop}} + t_{\text{tx}}(\text{ACK}) + t_{\text{prop}} \\
&= (N + 1) \cdot t_{\text{tx}} + 2 \cdot t_{\text{prop}} \\
&= 293 \times 2.8338 \mu\text{s} + 2 \times 84 \text{ ms} \\
&= 830.3 \mu\text{s} + 168 \text{ ms}
\end{aligned}$$

$$T_{\text{total}} \approx 168.8303 \text{ ms} \approx \mathbf{0.1688 \text{ s}}$$

Sources

- [1] Prof. Dr. Zsolt Istvan, “Computer Networks & Distributed Systems Chapter 4: Transport Layer.” 2026.
- [2] “DNS Checker.” [Online]. Available: <https://dnschecker.org/>